Integrated Inventory & Sales System

UML & Architecture Diagrams

Software Engineering Course — April 2026

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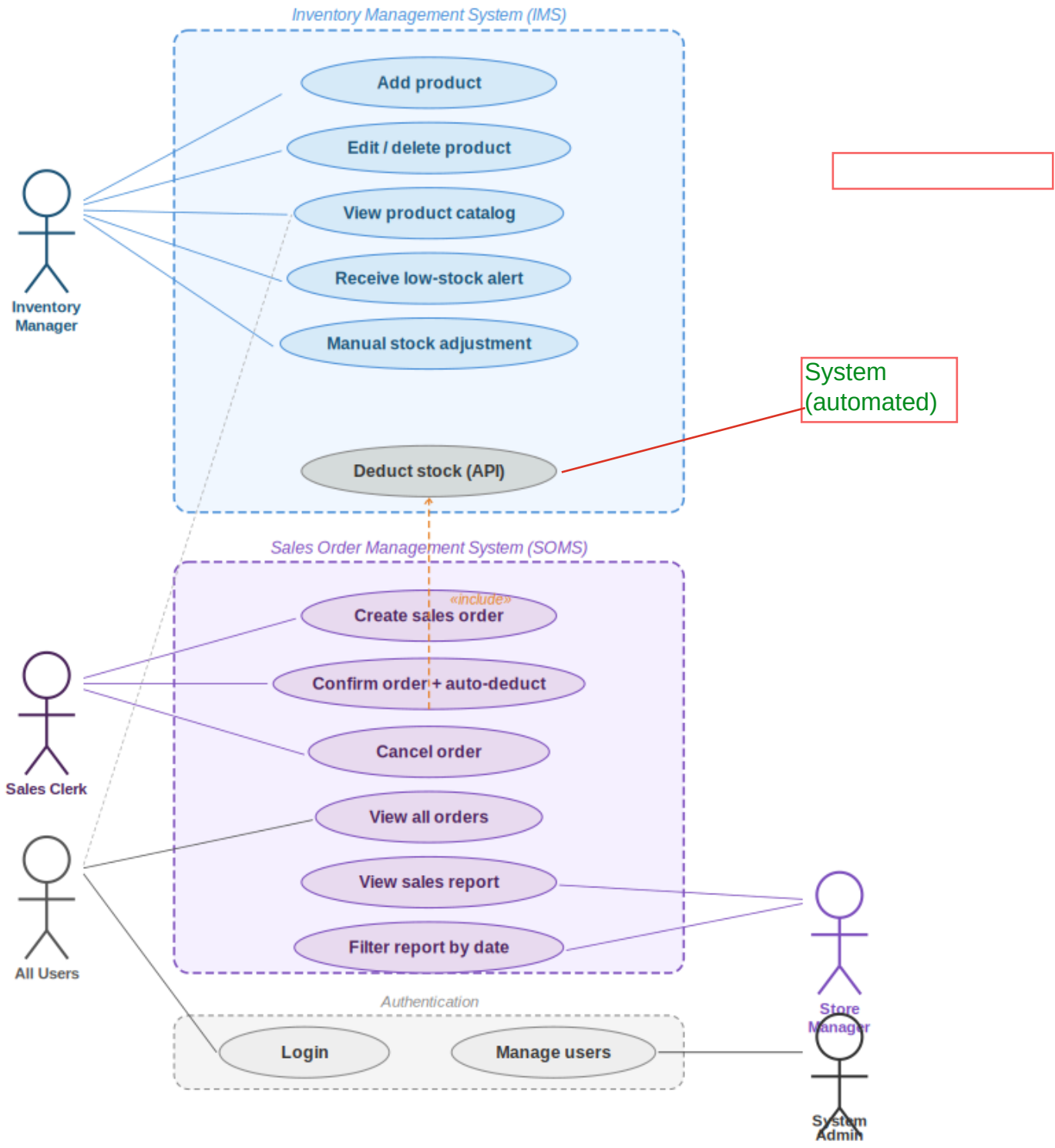
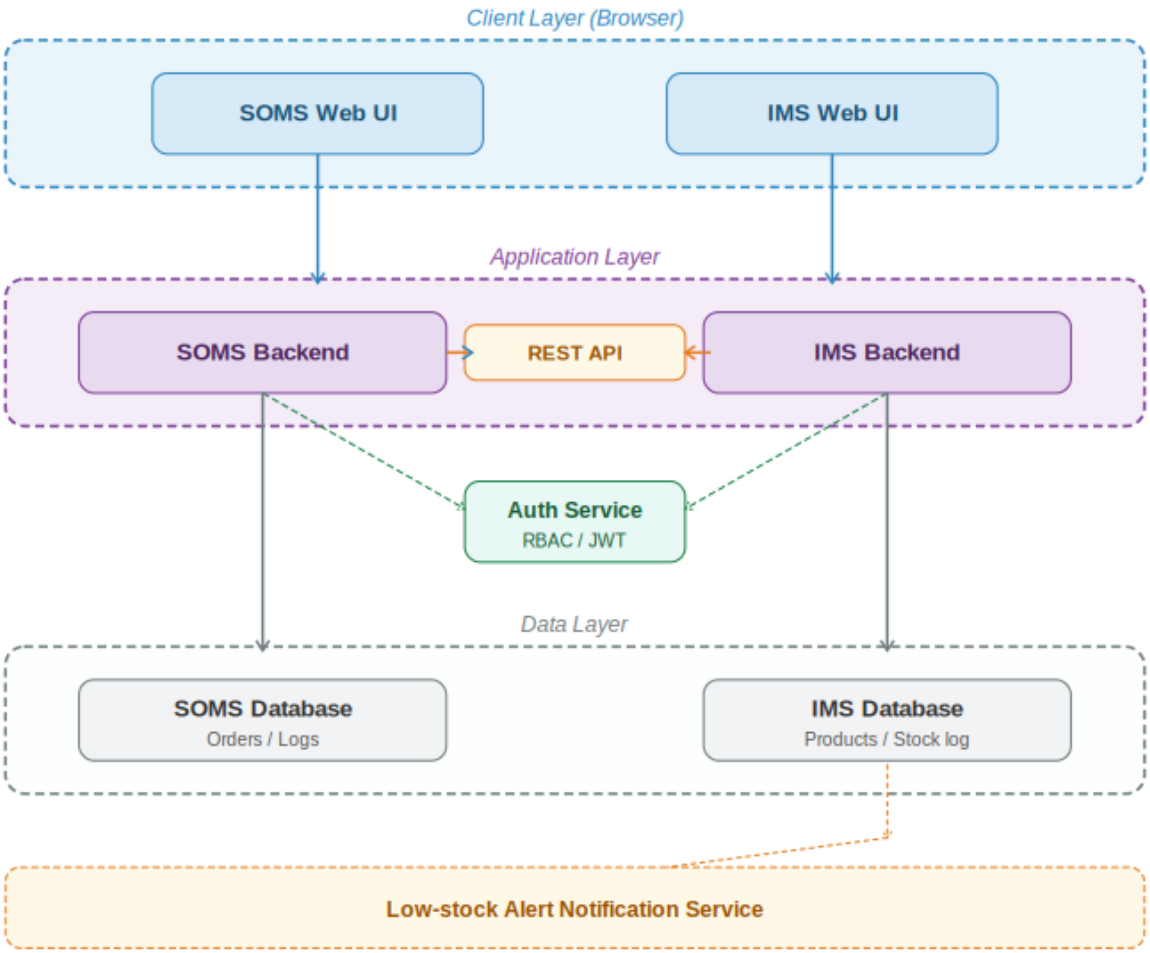
1. Use Case Diagram

Shows all actors (Inventory Manager, Sales Clerk, Store Manager, System Admin, All Users) and the 11 use cases they interact with across IMS and SOMS. The orange dashed arrow shows the <<include>> relationship between Confirm Order and Deduct Stock API.



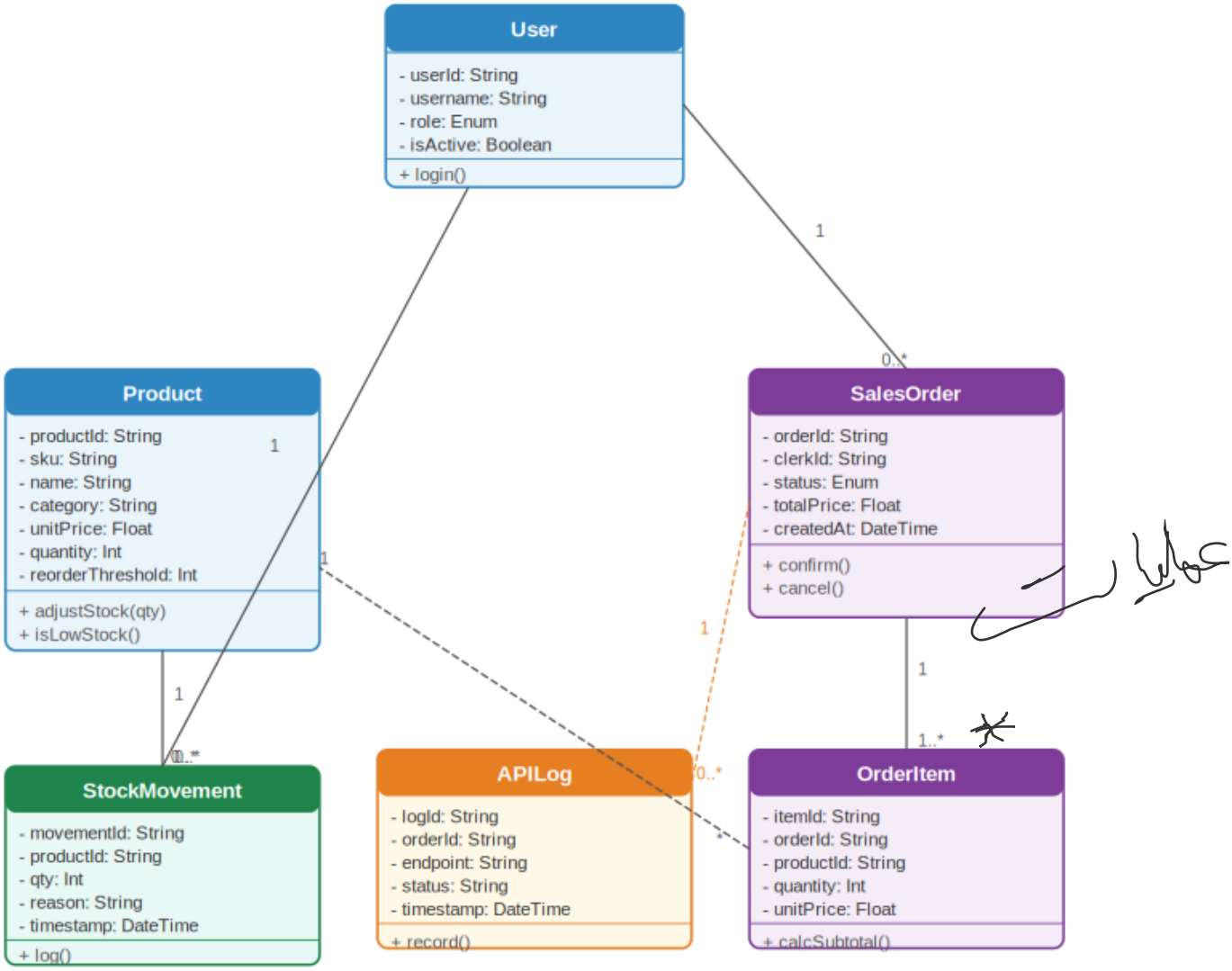
2. System Architecture Diagram

Three-tier architecture: Client Layer (browsers running SOMS & IMS web UIs), Application Layer (SOMS Backend and IMS Backend communicating via REST API, both secured by Auth Service with RBAC/JWT), and Data Layer (separate databases for SOMS and IMS). A Notification Service handles low-stock alerts.



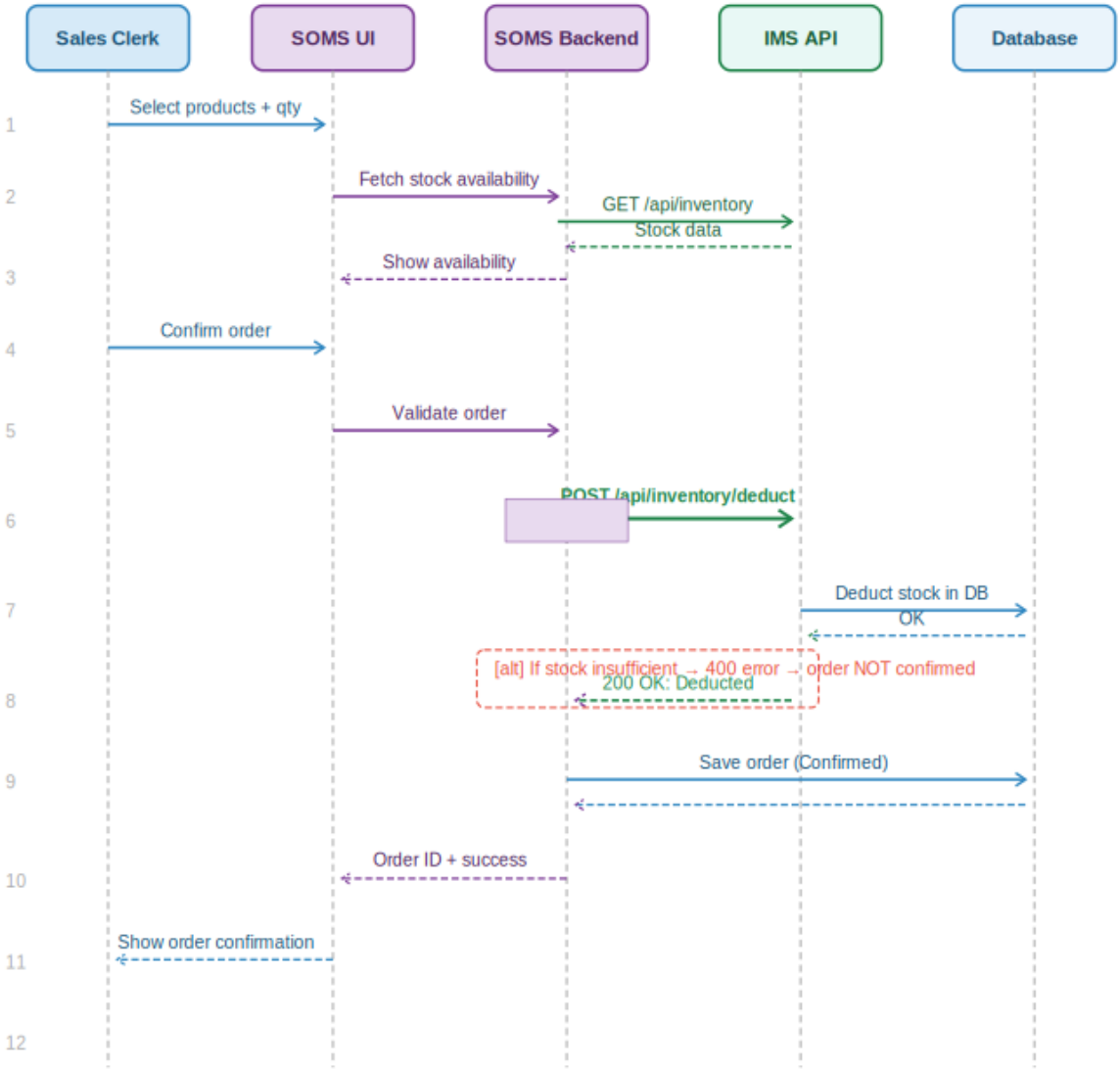
3. Class Diagram

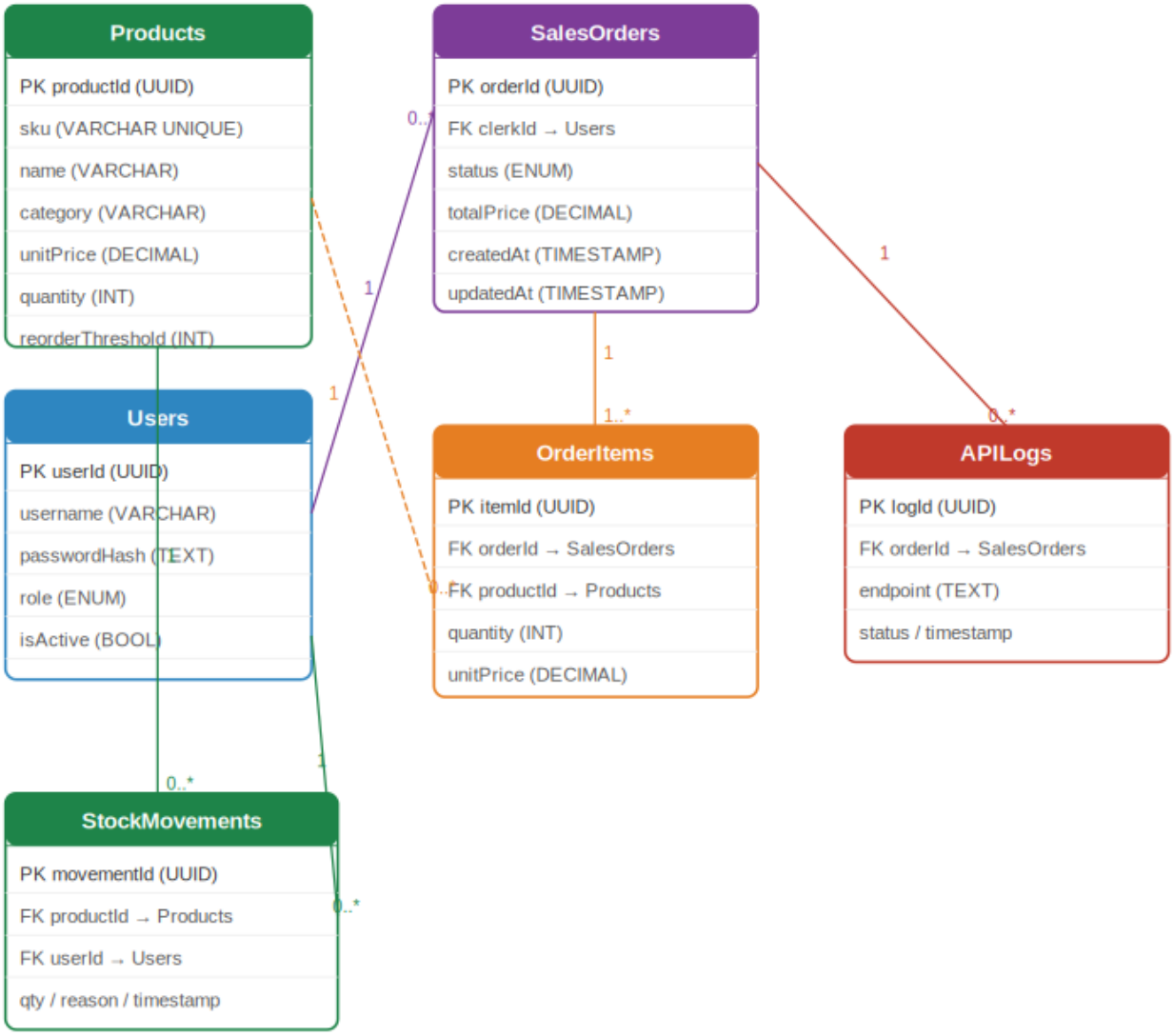
Six main classes: **User** (with role-based access), **Product** (catalog & stock), **SalesOrder** (order lifecycle), **OrderItem** (line items per order), **StockMovement** (audit log for all stock changes), and **APILog** (tracks every IMS API call from SOMS). Multiplicity relationships are shown on each association.



4. Sequence Diagram — Confirm Order Flow

Shows the 12-step interaction when a Sales Clerk confirms an order: from selecting products, fetching live stock from IMS, validating the order, calling POST /api/inventory/deduct, updating the database, and returning the order confirmation. The red alt fragment shows the failure path when stock is insufficient.



5. Entity Relationship Diagram (ERD)

Six database tables: **Users**, **Products**, **SalesOrders**, **OrderItems** (junction table linking orders to products), **StockMovements** (full audit log), and **APILogs** (records every IMS API call). Primary keys (PK) and foreign keys (FK) are marked on each table with their data types.